Incorrect Indication of Sensor/Line Malfunction at Shutdown Unit

Symptoms

- The SDU410 gives an alarm for wiring fault when no fault exists.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Check the SDU410 unit for the circuit giving the fault. If an open or short has occurred, follow the troubleshooting steps. If no trouble is found, disconnect the circuit and see if the fault goes away. If this removes the fault, then there is a problem with the intermittent fault circuit. If the fault is **not** removed then follow the troubleshooting steps.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check customer interface box	
	STEP 1A. Check the low speed oil pressure signal and return wires for an open.	Less than 10 ohms?
	STEP 1A-1. Check the low speed oil pressure signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 1A-2. Check the low speed oil pressure signal and return wires for a short to ground.	Less than 10 ohms?
	STEP 1B. Check the high speed oil pressure signal and return wires for an open.	Less than 10 ohms?

STEPS		SPECIFICATIONS
	STEP 1B-1. Check the high speed oil pressure signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 1B-2. Check the high speed oil pressure signal and return wires for a short to ground.	Less than 10 ohms?
	STEP 1C. Check the coolant pressure signal and return wires for an open.	Less than 10 ohms?
	STEP 1C-1. Check the coolant pressure signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 1C-2. Check the coolant pressure signal and return wires for a short to ground.	Less than 10 ohms?
	STEP 1D. Check the coolant temperature signal and return wires for an open.	Less than 10 ohms?
	STEP 1D-1. Check the coolant temperature signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 1D-2. Check the coolant temperature signal wire for a short to ground.	Less than 10 ohms?
	STEP 1E. Check the engine speed 1 signal and return wires for an open.	Less than 10 ohms?
	STEP 1E-1. Check the engine speed 1 signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 1E-2. Check the engine speed 1 signal and return wires for a short to ground.	Less than 10 ohms?
	STEP 1F. Check the engine speed 2 signal and return wires for an open.	Less than 10 ohms?

STEPS		SPECIFICATIONS
	STEP 1F-1. Check the engine speed 2 signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 1F-2. Check the engine speed 2 signal and return wires for a short to ground.	Less than 10 ohms?
	STEP 1G. Check the remote engine stop signal and return wires for an open.	Less than 10 ohms?
	STEP 1G-1. Check the remote engine stop signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 1G-2. Check the remote engine stop signal and return wires for a short to ground.	Less than 10 ohms?
	STEP 1H. Check the shutdown unit Modicon™ communication bus signal and return wires for an open.	Less than 10 ohms?
	STEP 1H-1. Check the shutdown unit Modicon™ communication bus signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 1H-2. Check the shutdown unit Modicon™ communication bus signal and return wires for a short to ground.	Less than 10 ohms?
	STEP 1I. Check the engine shutdown override signal and return wires for an open.	Less than 10 ohms?
	STEP 1I-1. Check the engine shutdown override signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 1I-2. Check the engine shutdown override signal and return wires for a short to ground.	Less than 10 ohms?
STEP 2.	Check the OEM wiring harness.	

STEPS		SPECIFICATIONS
	STEP 2A. Check the low speed oil pressure signal and return wires for an open.	Less than 10 ohms?
	STEP 2A-1. Check the low speed oil pressure signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 2B. Check the high speed oil pressure signal and return wires for an open.	Less than 10 ohms?
	STEP 2A-1. Check the high speed oil pressure signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 2C. Check the coolant pressure signal and return wires for an open.	Less than 10 ohms?
	STEP 2C-1. Check the coolant pressure signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 2D. Check the coolant temperature signal and return wires for an open.	Less than 10 ohms?
	STEP 2D-1. Check the coolant temperature signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 2E. Check the engine speed 1 signal and return wires for an open.	Less than 10 ohms?
	STEP 2E-1. Check the engine speed 1 signal and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 2F. Check the engine speed 2 signal and return wires for an open.	Less than 10 ohms?
	STEP 2F-1. Check the engine speed 2 signal and return wires for a wire-to-wire short.	Less than 10 ohms?

STEP 1. Check the customer interface box.

STEP 1A. Check the low speed oil pressure signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the low speed oil pressure signal and return wires at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step
Check the signal and return wires for an open. • Place one test lead on the low speed oil pressure signal wire at the SDU410 unit. • Place the other test lead on the low speed oil pressure signal pin at the C4 connector. • Place one test lead on the low speed oil pressure return wire at the SDU410 unit. • Place the other test lead on the low speed oil pressure return pin at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**1A-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 1A-1. Check the low speed oil pressure signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the low speed oil pressure signal and return wires at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step
Check the low speed oil pressure signal and return wires for a wire-to-wire short. • Place one test lead on the low speed oil pressure signal wire at the SDU410 unit. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the low speed oil pressure return pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1A-2**

STEP 1A-2. Check the low speed oil pressure signal and return wires for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the low speed oil pressure signal wire at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

Check the low speed oil pressure signal and return wires for a short to ground. • Place one test lead on the low speed oil pressure signal wire at the SDU410 unit. • Place the other test lead on panel ground. • Place one test lead on the low speed oil pressure return pin at the C4 connector. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1B**

STEP 1B. Check the high speed oil pressure signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the high speed oil pressure signal and return wires at the SDU410 unit and connector C4.

Action	Specification/Repair	Next Step

Check the high speed oil pressure signal and return wires for an open. • Place one test lead on the high speed oil pressure signal wire at the SDU410 unit. • Place the other test lead on the high speed oil pressure signal pin at the C4 connector. • Place one test lead on the high speed oil pressure return wire at the SDU410 unit. • Place the other test lead on the high speed oil pressure return pin at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**1B-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 1B-1. Check the high speed oil pressure signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the high speed oil pressure signal and return wires at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

Check the high speed oil pressure signal and return wires for a wire-to-wire short. • Place one test lead on the high speed oil pressure signal wire at the SDU410 unit. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the high speed oil pressure return pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1B-2**

STEP 1B-2. Check the high speed oil pressure signal and return wires for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the high speed oil pressure signal wire at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

Check the high speed oil pressure signal and return wires for a short to ground. • Place one test lead on the high speed oil pressure signal wire at the SDU410 unit. • Place the other test lead on panel ground. • Place one test lead on the high speed oil pressure return pin at the C4 connector. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1C**

STEP 1C. Check the coolant pressure signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the coolant pressure signal and return wires at the SDU410 unit and connector C4.

Action	Specification/Repair	Next Step

Check the coolant pressure signal and return wires for an open. • Place one test lead on the coolant pressure signal wire at the SDU410 unit. • Place the other test lead on the coolant pressure signal pin at the C4 connector. • Place one test lead on the coolant pressure return wire at the SDU410 unit. • Place the other test lead on the coolant pressure return pin at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**1C-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 1C-1. Check the coolant pressure signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the coolant pressure signal and return wires at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

<table>
<tr>
<td rowspan="2">Check the coolant pressure signal and return wires for a wire-to-wire short.
Place one test lead on the coolant pressure signal wire at the SDU410 unit.
Place the other test lead on all other pins at the C4 connector.
Place one test lead on the coolant pressure return pin at the C4 connector.
Place the other test lead on all other pins at the C4 connector.
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</td>
<td>Less than 10 ohms?
YES

Repair :
Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</td>
<td>Repair complete</td>
</tr>
<tr>
<td>Less than 10 ohms?
NO</td>
<td>1C-2</td>
</tr>
</table>

STEP 1C-2. Check the coolant pressure signal and return wires for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the coolant pressure signal wire at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

Check the coolant pressure signal and return wires for a short to ground. • Place one test lead on the coolant pressure signal wire at the SDU410 unit. • Place the other test lead on panel ground. • Place one test lead on the coolant pressure return pin at the C4 connector. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1D**

STEP 1D. Check the coolant temperature signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the coolant temperature signal and return wires at the SDU410 unit and connector C4.

Action	Specification/Repair	Next Step

Check the coolant temperature signal and return wires for an open. • Place one test lead on the coolant temperature signal wire at the SDU410 unit. • Place the other test lead on the coolant temperature signal pin at the C4 connector. • Place one test lead on the coolant temperature return wire at the SDU410 unit. • Place the other test lead on the coolant temperature return pin at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**1D-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 1D-1. Check the coolant temperature signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the coolant temperature signal and return wires at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

Check the coolant temperature signal and return wires for a wire-to-wire short. • Place one test lead on the coolant temperature signal wire at the SDU410 unit. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the coolant temperature return pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1D-2**

STEP 1D-2. Check the coolant temperature signal wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the coolant temperature signal wire at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

<table>
<tr><td rowspan="2">Check the coolant temperature signal wire for a short to ground.
Place one test lead on the coolant temperature signal wire at the SDU410 unit.
Place the other test lead on panel ground.
Place one test lead on the coolant temperature return pin at the C4 connector.
Place the other test lead on panel ground.
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</td><td>Less than 10 ohms?
YES

Repair :
Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</td><td>Repair complete</td></tr>
<tr><td>Less than 10 ohms?
NO</td><td>1E</td></tr>
</table>

STEP 1E. Check the engine speed 1 signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the engine speed 1 signal and return wires at the SDU410 unit and connector C4.

Action	Specification/Repair	Next Step

Check the engine speed 1 signal and return wires for an open. • Place one test lead on the engine speed 1 signal wire at the SDU410 unit. • Place the other test lead on the engine speed 1 signal pin at the C4 connector. • Place one test lead on the engine speed 1 return wire at the SDU410 unit. • Place the other test lead on the engine speed 1 return pin at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**1E-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 1E-1. Check the engine speed 1 signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the engine speed 1 signal and return wires at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

Check the engine speed 1 signal and return wires for a wire-to-wire short. • Place one test lead on the engine speed 1 signal wire at the SDU410 unit. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the engine speed 1 return pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1E-2**

STEP 1E-2. Check the engine speed 1 signal wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the engine speed 1 signal wire at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

Check the engine speed 1 signal wire for a short to ground. • Place one test lead on the engine speed 1 signal wire at the SDU410 unit. • Place the other test lead on panel ground. • Place one test lead on the engine speed 1 return pin at the C4 connector. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1F**

STEP 1F. Check the engine speed 2 signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the engine speed 2 signal and return wires at the SDU410 unit and connector C4.

Action	Specification/Repair	Next Step

<table>
<tr>
<td rowspan="2">Check the engine speed 2 signal and return wires for an open.
Place one test lead on the engine speed 2 signal wire at the SDU410 unit.
Place the other test lead on the engine speed 2 signal pin at the C4 connector.
Place one test lead on the engine speed 2 return wire at the SDU410 unit.
Place the other test lead on the engine speed 2 return pin at the C4 connector.
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</td>
<td>Less than 10 ohms?
YES</td>
<td>1F-1</td>
</tr>
<tr>
<td>Less than 10 ohms?
NO

Repair :
Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</td>
<td>Repair complete</td>
</tr>
</table>

STEP 1F-1. Check the engine speed 2 signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the engine speed 2 signal and return wires at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

<table>
<tr>
<td rowspan="2">Check the engine speed 2 signal and return wires for a wire-to-wire short.
Place one test lead on the engine speed 2 signal wire at the SDU410 unit.
Place the other test lead on all other pins at the C4 connector.
Place one test lead on the engine speed 2 return pin at the C4 connector.
Place the other test lead on all other pins at the C4 connector.
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</td>
<td>Less than 10 ohms?
YES
Repair :
Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</td>
<td>Repair complete</td>
</tr>
<tr>
<td>Less than 10 ohms?
NO</td>
<td>1F-2</td>
</tr>
</table>

STEP 1F-2. Check the engine speed 2 signal and return wires for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the engine speed 2 signal wire at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

Check the engine speed 2 signal and return wires for a short to ground. • Place one test lead on the engine speed 2 signal wire at the SDU410 unit. • Place the other test lead on panel ground. • Place one test lead on the engine speed 2 return pin at the C4 connector. • Place the other test lead on panel ground.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1G**

STEP 1G. Check the remote engine stop signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the remote engine stop signal and return wires at the SDU410 unit and connector X4.

Action	Specification/Repair	Next Step

Check the remote engine stop signal and return wires for an open. • Place one test lead on the remote engine stop signal wire at the SDU410 unit. • Place the other test lead on the remote engine stop signal wire at the X4 connector. • Place one test lead on the remote engine stop return wire at the SDU410 unit. • Place the other test lead on the remote engine stop return wire at the X4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**1G-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 1G-1. Check the remote engine stop signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the remote engine stop signal and return wires at the SDU410 unit and C4 connector.

Action	Specification/Repair	Next Step

Check the remote engine stop signal and return wires for a wire-to-wire short. • Place one test lead on the remote engine stop signal wire at the SDU410 unit. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the remote engine stop return pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1G-2**

STEP 1G-2. Check the remote engine stop signal and return wires for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the remote engine stop signal wire at the SDU410 unit and X4 connector.

Action	Specification/Repair	Next Step

Check the remote engine stop signal and return wires for a short to ground. • Place one test lead on the remote engine stop signal wire at the SDU410 unit. • Place the other test lead on panel ground. • Place one test lead on the remote engine stop return wire at the X4 connector. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1H**

STEP 1H. Check the shutdown unit Modicon™ communication bus supply and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply and return wires at the SDU410 unit and connector DCU410.

Action	Specification/Repair	Next Step

<table>
<tr>
<td rowspan="2">Check the shutdown unit Modicon™ communication bus supply and return wires for an open.

Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit.
Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the DCU410 unit.
Place one test lead on the shutdown unit Modicon™ communication bus return wire at the SDU410 unit.
Place the other test lead on the shutdown unit Modicon™ communication bus return wire at the DCU410 unit.

Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</td>
<td>Less than 10 ohms?
YES</td>
<td>1H-1</td>
</tr>
<tr>
<td>Less than 10 ohms?
NO

Repair :
Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</td>
<td>Repair complete</td>
</tr>
</table>

STEP 1H-1. Check the shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply and return wires at the SDU410 unit and DCU410 unit.

Action	Specification/Repair	Next Step

Check the shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short. • Place one test lead on the shutdown unit Modicon™ communication bus signal wire at the SDU410 unit. • Place the other test lead on all other wires at the DCU410 unit. • Place one test lead on the shutdown unit Modicon™ communication bus return wire at the SDU410 unit. • Place the other test lead on all other wires at the DCU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1H-2**

STEP 1H-2. Check the shutdown unit Modicon™ communication bus supply wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit and the DCU410 unit.

Action	Specification/Repair	Next Step

Check the shutdown unit Modicon™ communication bus supply wire for a short to ground. • Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit. • Place the other test lead on panel ground. • Place one test lead on the shutdown unit Modicon™ communication bus return pin at the C4 connector. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1I**

STEP 1I. Check the engine shutdown override signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the engine shutdown override signal and return wires at the SDU410 unit and connector DCU410.

Action	Specification/Repair	Next Step

Check the engine shutdown override signal and return wires for an open. • Place one test lead on the engine shutdown override signal wire at the SDU410 unit. • Place the other test lead on the engine shutdown override signal wire at the DCU410 unit. • Place one test lead on the engine shutdown override return wire at the SDU410 unit. • Place the other test lead on the engine shutdown override return wire at the DCU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**1I-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 1I-1. Check the engine shutdown override signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the engine shutdown override signal and return wires at the SDU410 unit and DCU410 unit.

Action	Specification/Repair	Next Step

Check the engine shutdown override signal and return wires for a wire-to-wire short. • Place one test lead on the engine shutdown override signal wire at the SDU410 unit. • Place the other test lead on all other wires at the DCU410 unit. • Place one test lead on the engine shutdown override return wire at the SDU410 unit. • Place the other test lead on all other wires at the DCU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**1I-2**

STEP 1I-2. Check the engine shutdown override signal wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the engine shutdown override signal wire at the SDU410 unit and DCU410 unit.

Action	Specification/Repair	Next Step

Check the engine shutdown override signal and return wires for a short to ground. • Place one test lead on the engine shutdown override signal wire at the SDU410 unit. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**2A**

STEP 2. Check the OEM wiring harness

STEP 2A. Check the low speed oil pressure signal and return wires for an open.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

<table>
<tr>
<td rowspan="2">Check the low speed oil pressure signal and return wires for an open.
Place one test lead on the low speed oil pressure signal pin at the C4 connector.
Place the other test lead on the low speed oil pressure signal pin at the C11 connector.
Place one test lead on the low speed oil pressure return pin at the C4 connector.
Place the other test lead on the low speed oil pressure return pin at the C11 connector.
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</td>
<td>Less than 10 ohms?
YES</td>
<td>2A-1</td>
</tr>
<tr>
<td>Less than 10 ohms?
NO
Repair :
Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</td>
<td>Repair complete</td>
</tr>
</table>

STEP 2A-1. Check the low speed oil pressure signal and return wires for a wire-to-wire short.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

Check the low speed oil pressure signal and return wires for a wire-to-wire short. • Place one test lead on the low speed oil pressure signal pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the low speed oil pressure return pin at the C11 connector. • Place the other test lead on all other pins at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**2B**

STEP 2B. Check the high speed oil pressure signal and return wires for an open.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

Check the high speed oil pressure signal and return wires for an open. • Place one test lead on the high speed oil pressure signal pin at the C4 connector. • Place the other test lead on the high speed oil pressure signal pin at the C11 connector. • Place one test lead on the high speed oil pressure return pin at the C4 connector. • Place the other test lead on the high speed oil pressure return pin at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**2B-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 2B-1. Check the high speed oil pressure signal and return wires for a wire-to-wire short.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

Check the high speed oil pressure signal and return wires for a wire-to-wire short. • Place one test lead on the high speed oil pressure signal pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the high speed oil pressure return pin at the C11 connector. • Place the other test lead on all other pins at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**2C**

STEP 2C. Check the coolant pressure signal and return wires for an open.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

Check the coolant pressure signal and return wires for an open. • Place one test lead on the coolant pressure signal pin at the C4 connector. • Place the other test lead on the coolant pressure signal pin at the C11 connector. • Place one test lead on the coolant pressure return pin at the C4 connector. • Place the other test lead on the coolant pressure return pin at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**2C-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 2C-1. Check the coolant pressure signal and return wires for a wire-to-wire short.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

<table>
<tr><td rowspan="2">Check the coolant pressure signal and return wires for a wire-to-wire short.
Place one test lead on the coolant pressure signal pin at the C4 connector.
Place the other test lead on all other pins at the C4 connector.
Place one test lead on the coolant pressure return pin at the C11 connector.
Place the other test lead on all other pins at the C11 connector.
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</td><td>Less than 10 ohms?
YES
Repair :
Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</td><td>Repair complete</td></tr>
<tr><td>Less than 10 ohms?
NO</td><td>2D</td></tr>
</table>

STEP 2D. Check the coolant temperature signal and return wires for an open.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

Check the coolant temperature signal and return wires for an open. • Place one test lead on the coolant temperature signal pin at the C4 connector. • Place the other test lead on the coolant temperature signal pin at the C11 connector. • Place one test lead on the coolant temperature return pin at the C4 connector. • Place the other test lead on the coolant temperature return pin at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**2D-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 2D-1. Check the coolant temperature signal and return wires for a wire-to-wire short.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

Check the coolant temperature signal and return wires for a wire-to-wire short. • Place one test lead on the coolant temperature signal pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the coolant temperature return pin at the C11 connector. • Place the other test lead on all other pins at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**2E**

STEP 2E. Check the engine speed 1 signal and return wires for an open.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

Check the engine speed 1 signal and return wires for an open. • Place one test lead on the engine speed 1 signal pin at the C4 connector. • Place the other test lead on the engine speed 1 signal pin at the C11 connector. • Place one test lead on the engine speed 1 return pin at the C4 connector. • Place the other test lead on the engine speed 1 return pin at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**2E-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 2E-1. Check the engine speed 1 signal and return wires for a wire-to-wire short.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

Check the engine speed 1 signal and return wires for a wire-to-wire short. • Place one test lead on the engine speed 1 signal pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the engine speed 1 return pin at the C11 connector. • Place the other test lead on all other pins at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**2F**

STEP 2F. Check the engine speed 2 signal and return wires for an open.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

Check the engine speed 2 signal and return wires for an open. • Place one test lead on the engine speed 2 signal pin at the C4 connector. • Place the other test lead on the engine speed 2 signal pin at the C11 connector. • Place one test lead on the engine speed 2 return pin at the C4 connector. • Place the other test lead on the engine speed 2 return pin at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES**	**2F-1**
	Less than 10 ohms? **NO** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**

STEP 2F-1. Check the engine speed 2 signal and return wires for a wire-to-wire short.

Conditions :

- Disconnect the OEM wiring harness from the customer interface box at the C4 connector.
- Disconnect the OEM wiring harness at the C11 connector.

Action	Specification/Repair	Next Step

Check the engine speed 2 signal and return wires for a wire-to-wire short. • Place one test lead on the engine speed 2 signal pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the engine speed 2 return pin at the C11 connector. • Place the other test lead on all other pins at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? **YES** **Repair :** Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	**Repair complete**
	Less than 10 ohms? **NO**	**Troubleshooting procedures must be checked again from the beginning. A fault mode should have been detected.**

Last Modified: 04-Apr-2008